TEMA 2. Ètica, per què?

"Tecnoètica" (M. Bunge)

TOWARDS A TECHNOETHICS *

Gone are the days of the divine right of kings—or of anyone else, whether owner, manager, labor leader, politician, bureaucrat, technologist or scholar. Absolute and groundless authority is being contested all over the world: ours is an iconoclastic age. Moreover nobody recognizes rights without duties, privileges without responsibilities. Everyone is rightly held responsible for what he does and even for what he fails to do when he ought to act. And the responsibilities are not to some conveniently distant deity or sovereign, or even to the anonymous people, but to definite individuals—superiors, peers, subordinates, neighbors, the public, and even possible future humans.

Moreover the old separations among different kinds of responsibility—moral, professional, social, etc.—are being lifted. We are beginning to realize that the separation of responsibilities is just a retreat from total responsibility, hence a cloak for wrongdoing. A person in charge of something, be it a machine or another individual, is not composed of a number of moral entities but is a single person, now acting in one capacity, now in another. All these various capacities should combine harmoniously. Being an affectionate parent does not exculpate any crimes; being a competent engineer does not confer rights of piracy on the environment; being an efficient manager does not entitle him to oppress others. Every human being has a number of intertwined responsibilities and each of them is as personal and intransferable as a joy or a grief.

This paper examines some of the special responsibilities of the technologist in our age of pervasive—and, alas, all too often perverse—technology. We shall defend the thesis that the technologist, just like anyone else, is personally responsible for whatever he does, and that he is responsible to all mankind not just to his employers. We shall also claim that the technologist has the duty to face, ponder over, and solve his own moral problems. And we shall submit that he is singularly privileged to do so, as he can tackle moral problems, and even the theory of morality, i.e. ethics, with the help of an approach and a set of tools alien to most philosophers and yet promising to deliver the technoethics that philosophers have not deigned to work on. To this end we shall put forth a value theory allowing one to weigh

*Paper discussed at the International Symposium on Ethics in an Age of Pervasive Technology held at the Technion-Israel Institute of Technology, December 21–25, 1974.

means and ends, and to conceive of moral norms in the image of technological rules.

1. The Inescapable Responsibilities of the Technologist

Look around and you will easily recognize the professions that have most contributed to shaping industrial society. They are the scientists, the engineers, and the managers (including the politicians). The former have supplied the basic knowledge, the engineers have used the latter to design their works, and the managers have organized the manpower that has implemented those designs. The outcome of their labors is well known: a new kind of society, one that may carry mankind either to a higher evolutionary level or to quick extinction.

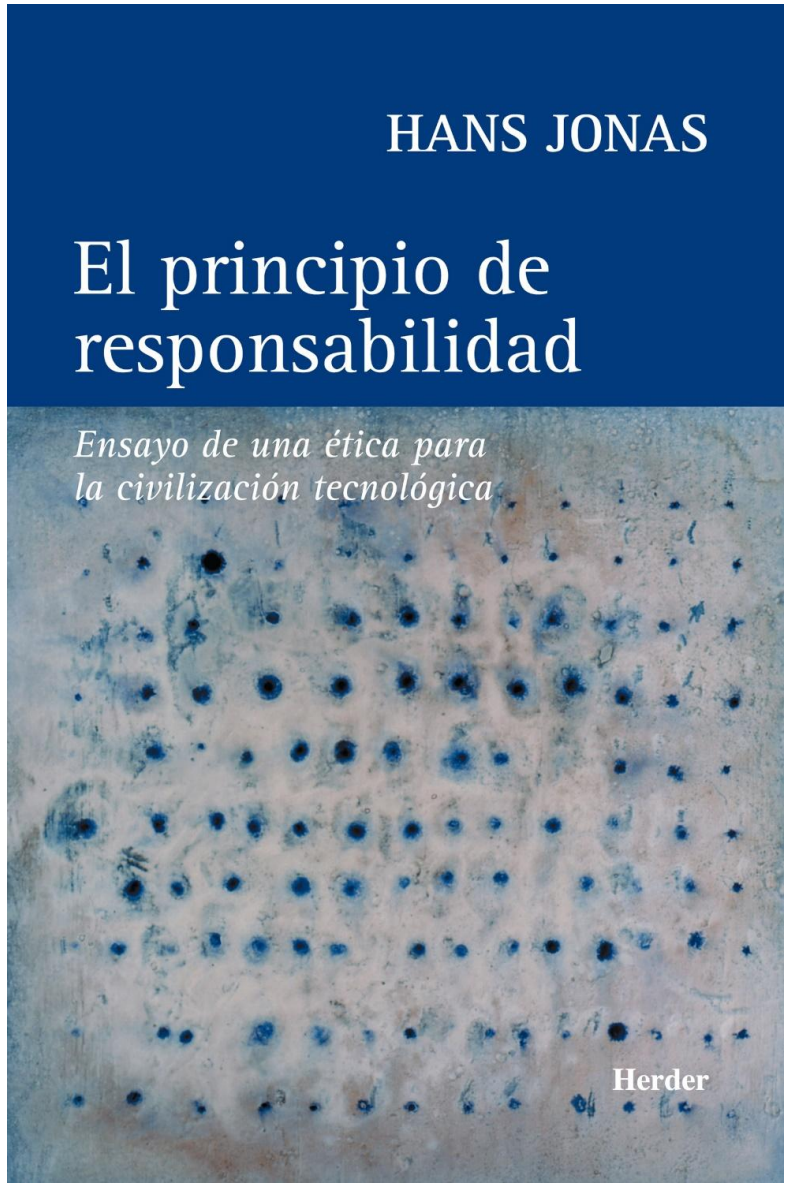
And yet, paradoxically enough, the sociologist tells us that, by and large, scientists, engineers and managers—that is, the main artificers of modern society—do not feel constrained or fired by any extraprofessional moral and social responsibilities. In particular the technologists seem to be indifferent, when not outright callous, in the face of large-scale yet avoidable tragedy—unemployment, poverty, inequity, oppression, war, the mutilation of nature, the squandering of natural resources, or the debasement of culture.

Whether or not they have a hand in eliciting any calamities, most scientists, engineers and managers disclaim any responsibilities and shut their eyes to suffering and squalor. Worse, their position in society is such that they must play blind if they wish to perform efficiently. In fact no professional can do his job efficiently if distracted and upset by cries of distress: he must shut himself up in his office or laboratory if he is to get on with his specific work, be it investigating, designing, or organizing. Unless of course it is part of his job to relieve that very distress or at least avert future sources of distress.

True, some professional bodies have imitated the physicians and adopted moral codes ruling their own work. But most of these codes are limited to professional responsibilities, so they leave ample margin for irresponsibility. The minutiae are taken care of, the great issues pushed aside. Hence the scientist feels free to go on with his research come what may; the engineer, to go ahead with his projects go what or who may; and the manager, to push production or sales come or go what or who may.

Thus there are hardly any external constraints capable of being internalized and preventing the scientist, the engineer and the manager from engaging in morally wrong or socially injurious activities qua professionals. Whether or not he behaves well towards his fellow human beings is left to his moral conscience or, worse, to that of his superiors. Unfortunately conscience, usually wakeful in private and professional matters, is rather sleepy when it comes to affecting the anonymous lives of others.

Ètica de la responsabilitat (H. Jonas)



Punts centrals del plantejament de l'ètica de la responsabilitat

I. El caràcter modificat de l'acció humana

II. L' *homo faber* per sobre de l' *homo sapiens*



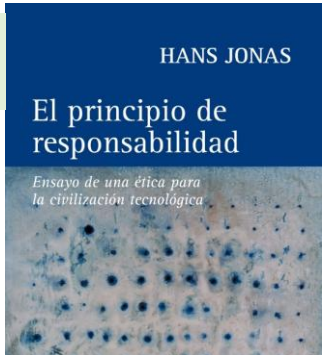
III. Característiques (i insuficiències) de les ètiques anteriors

IV. «Només sabem ***què*** està en joc quan sabem ***que*** està en joc»

V. Heurística del temor com a mètode d'aproximació ètica als riscos

VI. El principi de responsabilitat:
«Obra de tal manera que els efectes de la teva acció siguin compatibles amb la permanència d'una vida humana autèntica a la Terra»

VII. Antropologia de l'ètica de la responsabilitat
«...només la prevista *desfiguració* de l'home ens ajuda a forjar-nos la idea d'home que ha de ser *preservada* de tal desfiguració»



EL BIEN, EL DEBER Y EL SER 203

deseo y por la necesidad y alimentadas a menudo por la fe supersticiosa en la omnipotencia de la ciencia. Tras lo que hemos visto en el último siglo no podemos excluir, por ejemplo, que se descubran nuevas fuentes de energía o formas totalmente diferentes de las ya conocidas; tampoco podemos excluir la posibilidad de que las sorpresas agradables del progreso no tengan fin y que unas u otras nos saquen de apuros a tiempo. Pero sería completamente irresponsable edificar nuestro futuro sobre tales expectativas[24]. Igualmente irresponsable sería edificarlo sobre la predicción (por muchas probabilidades que *esa* predicción tenga de ser cierta) de que el hombre puede adaptarse a cualquier cosa. De hecho, si la vida significa fundamentalmente adaptación, éste será el mejor y más eficaz seguro de vida que podrán ofrecernos los apóstoles de la incesante transformación tecnológica de las condiciones de la existencia. Nosotros afirmamos que asentar el futuro sobre tal certeza (una vez se haya dado por buena) es, por lo menos, tan irresponsable como lo era en el anterior grupo de ejemplos el abandono a lo incierto. Pues la pregunta no es aquí «¿se logrará?» (hay que temer que sí), sino «¿a qué le es *lícito* al hombre habituarse?», «¿a qué es lícito forzarle o permitirle que se habitúe?», «¿qué condiciones de adaptación es lícito permitir?». Estas cuestiones ponen en juego la idea de hombre. También ella forma parte de la responsabilidad del gobernante, es su contenido último y a la vez más próximo, el núcleo de su totalidad, el auténtico horizonte de su apertura al futuro. Más tarde hablaremos de ello.

(1) Identificació

Què és un risc?
Què compta com a risc?
Quin risc?
Per a qui, és un risc?

Ignorància i incertesa → coordenades del **Principi de precaució**

(2) Quan és moralment acceptable, un risc?



La permissibilitat moral d'una activitat tecnològica té a veure *exclusivament* amb els riscos d'aquesta?

Recull d'objeccions ètiques a la clonació humana

Cloning:

1 undermines the uniqueness of humans;
2 is contrary to human dignity;
3 leads to psychosocial problems in the cloned child;
4 suffers from numerous scientific and technical risks;
5 will lead to misuse and has unforeseen and undesirable consequences;
6 is unnatural;
7 is based on an overreaching desire for manipulation and leads to the abject instrumentalization of people;
8 will lead to undesirable changes in our self-image.

Source: Based on Swierstra (2000, p. 42).

(1) Quines d'aquestes objeccions a la clonació humana es poden formular en termes de risc, i quines no?

(2) Quins són els models ètics que sustenten els arguments no conseqüencialistes contra la clonació humana?

(3) Són totes raonables, les objeccions formulades?

✓ L'acceptabilitat moral d'un risc

4 consideracions:

1. Consentiment informat dels afectats
2. Excés de beneficis per sobre dels riscos
   ! Alerta amb la *pricing fallacy*
3. Disponibilitat d'alternatives amb riscos menors
4. Distribució justa de costos-beneficis

(3) Com gestionar el risc?

Comunicació pública del risc → **quina** informació?
→ **com** comunicar?

Risk Perception

From psychological research on risk perception among the public, it appears a large number of factors plays a role on how people perceive risks:[11]

- The (perceived) voluntariness of the risk;
- The expected benefits of the activity or technology that causes the risk and the distribution of these effects;
- The maximal occurring negative effects and the possible controllability of these effects (degree of social disruption);
- The situation the risk is related to. Many people estimate risks related to work as smaller compared to other situations;
- The proximity and visibility of risks. Close risks are usually experienced as being greater than risks further away, or those that are less visible or imaginable;
- The way risks are presented and the risk measure that is used.

Source: Based on Slovic, Fischhoff, and Lichtenstein (1990) and Martin and Schinzinger (1996, pp. 134–137).

Exemple:

- Els costos de l'entrenament d'un model algorítmic són de 626.000 lliures de CO_2; això és, **284 tones mètriques de CO_2**)

vs.

- L'emissió de **CO_2** per l'entrenament d'un model algorítmic és **5 vegades superior a les emissions de la vida útil d'un cotxe**, o casi equivalents a **2.800 vols** Madrid-BCN, o equivalents al que contaminen **45 persones** europees en un any.

El cas del Ford Pinto

The societal benefits of the riskier design that costs $11 less was estimated at nearly $50 million: 180 lives lost, 180 wounded and 2100 cars burnt out. The calculation for this was 180 lives × $200,000 + 180 seriously wounded × $67,000 + 2100 burnt out cars × $700 = $49.53 million. This was considered to be the total societal benefit.

Against this there was the cost of improving the cars: 11 million cars and 1.5 million trucks had to be called back and retrofitted against an estimated costs per unit of $11, amounting to a total cost of 137 million dollar (12.5 million × $11). A memorandum attached to the report described the costs and benefits as in Table 3.1.

Table 3.1 Benefits and costs

Benefits		
Savings	*Unit cost (US$)*	*Total (US$)*
180 burn deaths	200 000	36 000 000
180 serious burn injuries	67 000	12 060 000
2,100 burned vehicles	700	1 470 000
Total		*49 530 000*
Costs		
Sales	*Unit cost (US$)*	*Total (US$)*
11 million cars	11 per car	121 000 000
1.5 million light trucks	11 per truck	16 500 000
Total		*137 500 000*

The estimation by Ford of the number of lives lost and wounded incurred was based on statistical data. The estimation Ford made that a human life is worth $200,000 was based on a report of the National Highway Traffic Safety Administration (see Table 3.2).

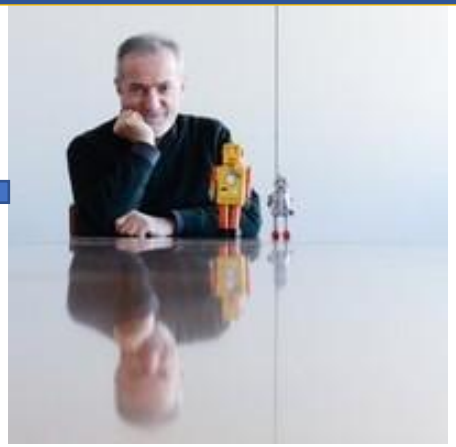
Table 3.2 Component costs

Component		*1971 costs (US$)*
Future productivity losses	Direct	132 000
	Indirect	41 300
Medical costs	Hospital	700
	Other	425
Property damage		1 500
Insurance administration		4 700
Legal and Court		3 000
Employer losses		1 000
Victim's pain and suffering		10 000
Funeral		900
Assets (lost consumption)		5 000
Miscellaneous accident cost		200
Total per fatality		*200 725*

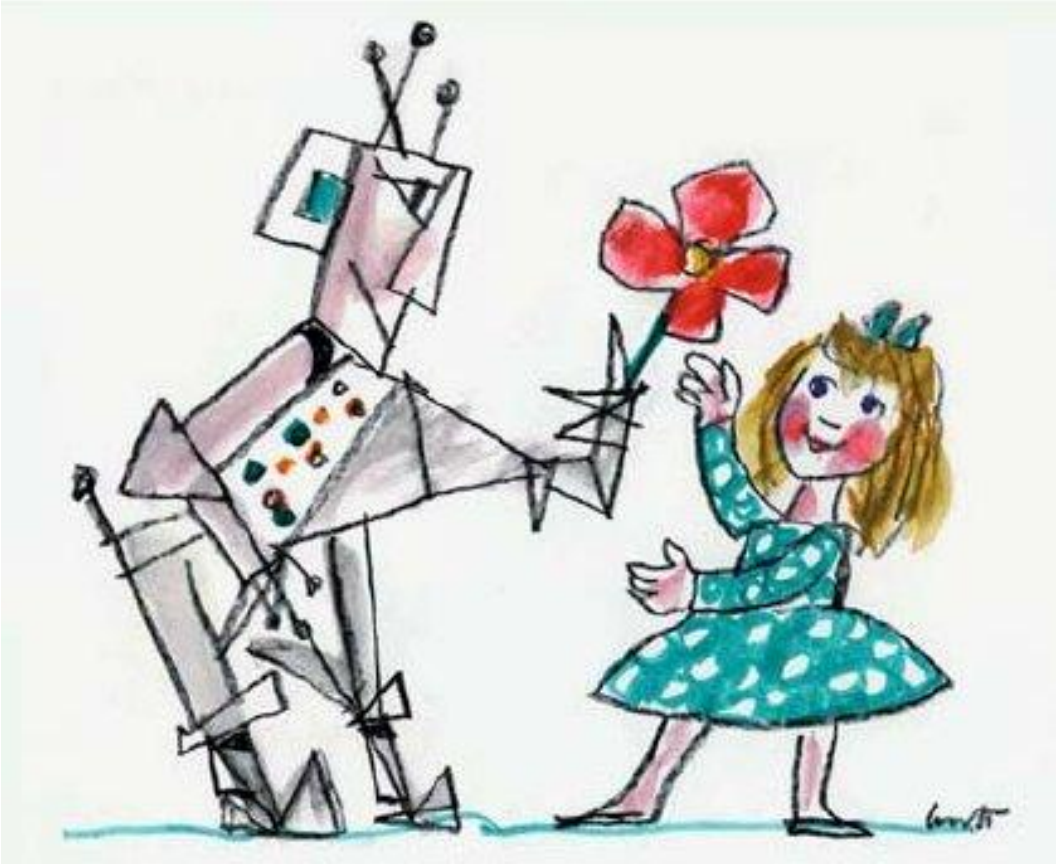
The conclusion that Ford drew was clear: a technical improvement costing $11 per car which would have prevented gas tanks from rupturing so easily was not cost-effective for society. The $137 million cost of the safer model clearly outweighed the benefits of $49.53 million. Altering the Pinto for $11 a car would cost society ($137 million – $49.53 million) $87.47 million.

(1) És pertinent l'avaluació de l'acceptabilitat moral dels riscos feta per Ford? Quines condicions per a l'avaluació ètica dels riscos va complir i quines no?

(2) Com valoreu la gestió del risc que va fer Ford? Creieu que s'hagués hagut de fer d'una altra manera? Com i per què?



ROBOÈTICA (2004)



DISCUSSIÓ (a): Una moratòria per a la IA?

https://futureoflife.org/open-letter/pause-giant-ai-experiments/

Pause Giant AI Experiments: An Open Letter

We call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4.

(1) Quines són les **raons** que es donen per a aturar l'entrenament/desenvolupament de models de llenguatge d'IA?

(2) Quin és el **model ètic** predominant de l'argumentació a favor de la moratòria?

(3) Quin és el **principi ètic principal** des del qual s'articula la defensa de la moratòria?
A quin **plantejament històric** de l'ètica de la tecnologia diries que està vinculat (tesis, autor)?

(4) Considereu que l'argumentació ètica en defensa de la moratòria és **raonable** (bona)? **Per què?**
Se li podria fer alguna refutació important?

(5) Creieu que la mesura de la moratòria és, almenys en cert grau, una **bona gestió** del problema a què respon? **Per què?**

DISCUSSIÓ (b): Una moratòria per a la IA?

◊ Anàlisi article d'argumentació sobre la moratòria a la IA

"Una moratoria artificial" (D. Innerarity)

(1) Quantes i quines **raons** ofereix D. Innerarity contra la petició d'una moratòria a la IA?

(2) Per a què sí que és una **bona mesura**, la Carta Oberta per la moratòria, segons D. Innerarity?

(3) Quina és la **narrativa/imaginari sobre la IA** que Innerarity identifica en la Carta Oberta, i quina **valoració** en fa?

(4) Quin considereu que és **l'argument menys convincent** dels que formula l'autor contra la petició de moratòria a la IA? Com el **contra argumentaríeu**?